Marine Oil-Spill Detection, Source Attribution & Trajectory Prediction

Literature Review, Technology Survey & System Design

Scope note on methodology: This review is built from primary sources (peer-reviewed journals, preprints, official agency pages, dataset/API documentation, and GitHub repositories) retrieved and read directly for this report. It is a **representative, source-grounded survey**, not an exhaustive census of the field (the literature runs into hundreds of papers). Every figure below is either taken from a cited source or marked **"not reported"** — nothing is invented. Where I could not verify a number (e.g., exact dataset size or a specific accuracy figure) from the sources retrieved, I say so explicitly rather than estimate. You should treat this as a strong starting map; before final engineering decisions, verify load-bearing numbers (accuracy claims, licensing terms, API limits) against the primary source link given.

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1. Category A — Satellite-Based Oil-Spill Detection (general)

#	Title / System	Year	Org / Authors	Region	Sensor(s)	Method	Key metric	Public?	Link
A1	Oil Spill Identification from Satellite Images Using Deep Neural Networks (the "Krestenitis" benchmark paper)	2019	Krestenitis, Orfanidis, Ioannidis, Avgerinakis, Vrochidis, Kompatsiaris — CERTH/MKLab, Greece	Global (EMSA-sourced)	Sentinel-1 SAR (VV)	U-Net, LinkNet, PSPNet, DeepLabv2, DeepLabv3+ compared	DeepLabv3+ best overall; U-Net/DeepLabv3+ IoU on oil-spill class ≈53.8%/53.4% (as reported by a later comparative study)	Dataset available on request from MKLab/CERTH (not fully open-download)	DOI
A2	Large-Scale Detection and Categorization of Oil Spills from SAR Images with Deep Learning	2020	KSAT-affiliated researchers	Global (North Sea/Barents Sea emphasis)	Sentinel-1	Segmentation CNN + shape/texture categorization	713 products, 2,093 oil-spill events, 2014–2018, ~150,000 km²/scene	Paper open (arXiv); underlying KSAT operational dataset not public	arXiv
A3	Automated oil spill detection using deep learning and SAR data — Suez Canal	2025	Multiple authors, Egypt	Suez Canal, Egypt	Sentinel-1	DeepLabv3+ (EfficientNet backbone)	Accuracy 98.14%, MIoU 0.7872 (Egyptian-data model) vs EMSA-CSN-trained model (higher accuracy, lower MIoU)	Code/data not stated as public	Nature Sci Rep
A4	Marine oil spill detection and segmentation — two-step DL framework	2024	Trujillo-Acatitla et al.	Mexico (new Sentinel-1 dataset built)	Sentinel-1	CNN (classification) + U-Net (segmentation, Focal Loss)	Classification 99% accuracy; segmentation 99% accuracy, 96% IoU	Not reported as public	PubMed
A5	An affordable operational oil-spill monitoring system — Gulf of Mexico	2024	Multi-institution	Southern Gulf of Mexico / Campeche Bank	Multi-satellite (SAR + optical)	Operational visual/algorithmic detection + drifter/HF-radar validation	>1,500 images analyzed, 70+ warning reports, spills tracked 12–24 days, up to 669 km²	Operational system, not open-source	PMC
A6	A near real-time automated oil-spill detection & early-warning system — SE	2024	Academic team	SE Mediterranean	Sentinel-1	Semantic segmentation pipeline, discusses operational compute cost of full-scene pixel	Not reported	Not stated	Tandfonline

	Mediterranean					classification			
A7	A Review of Oil Spill Detection Using Satellite Remote Sensing & Google Earth Engine	2026	Academic review	Global	Optical + SAR	Review of GEE-based pipelines for scalable detection	Review paper; no single metric	Review is open access	MDPI

Take-away for your app: the field has converged on Sentinel-1 SAR + semantic segmentation (U-Net/DeepLabv3+ family) as the dominant approach, largely because Sentinel-1 is free, all-weather, and day/night-capable, whereas optical (Sentinel-2) is a secondary confirmation sensor.

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2. Category B — SAR Segmentation & Deep-Learning Models

#	Study	Year	Model(s)	Dataset	Reported metric	Notes
B1	Krestenitis et al. (benchmark)	2019	U-Net, LinkNet, PSPNet, DeepLabv2, DeepLabv3+	1,112 Sentinel-1 images, 5 classes (sea, oil, look-alike, ship, land)	DeepLabv3+ outperformed others on boundary quality; U-Net/DeepLabv3+ oil-class IoU ≈53.8%/53.4% per a later comparative citation	The **de facto standard benchmark**; 280+ citations; distributed on request only
B2	Ma et al. — improved DeepLabv3+	2022	Modified DeepLabv3+	35 Sentinel-1 scenes	Oil-spill IoU 91.58%	Small, curated dataset — small-N results are not directly comparable to large-N studies
B3	Zhu et al. — CBD-Net	2022	Custom CNN ("CBD-Net")	14 ALOS PALSAR + 7 Sentinel-1 scenes	Precision 84.2% (PALSAR) / 91.2% (Sentinel-1)	Cross-sensor comparison
B4	Dual-Attention U-Net	2022	U-Net + Dual Attention Module (DAM)	Krestenitis benchmark (1,112 images)	Improvement over vanilla U-Net (exact IoU not extracted from source)	Published in Springer *J. Indian Society of Remote Sensing*
B5	Few-shot oil-spill segmentation, multi-scale feature similarity	2024	Few-shot segmentation network	Krestenitis-derived: 1,002 train / 110 test, 10 m resolution	Not reported in retrieved excerpt	Targets small-sample regimes — relevant if your labeled data is scarce
B6	ViR-SC hybrid ensemble (denoising autoencoder + U-Net++ + ResNet18 + ViT + RF + SVM)	2026	Ensemble: CNN, ResNet18, Vision Transformer, Random Forest, SVM + U-Net++ segmentation	Sentinel-1 (150 test patches for denoiser eval)	IoU 0.78, Dice 0.86, pixel accuracy 93.5%, mean IoU 0.74, precision/recall 0.88/0.85, Boundary F1 0.72	Notable for combining DL + classical ML + explainability (SHAP-style) in one pipeline — a good architectural template
B7	Compositional detector + adapted Segment Anything Model (SAM)	2024	Object detector + SAM adaptation	Not fully specified in excerpt	Not reported	Notes that ~90% of slicks lie within 160 km of shorelines and slick belts resemble shipping routes — directly useful prior for your candidate-vessel search
B8	SAR Oil Spill Detection via Random Forest classifiers (open-source)	2021	Two Random Forest classifiers	Sentinel-1	Not reported (numeric)	Fully open-source code + trained models, explicitly designed to reject look-alikes; good classical-ML baseline / fallback for low-compute deployments

Model landscape summary: DeepLabv3+ (atrous conv + ASPP + encoder-decoder) is reported as the strongest general segmentation backbone across most studies; U-Net variants remain the most widely used baseline and are easiest to adapt to small labeled datasets; Mask R-CNN appears mainly where instance-level (per-slick) detection matters, not just pixel labeling.

3. Category C — Look-Alike Discrimination

Oil spills and "look-alikes" both dampen capillary waves and appear as dark patches in SAR — the central operational challenge.

#	Study	Year	Look-alike classes addressed	Method	Reported result
C1	Compact-polarimetric SAR + feature-enriched Random Forest	2025	Biogenic films, low wind, rain cells, shear zones, internal waves, ship wakes	Polarimetric feature extraction + RF	Prior work cited: biogenic-class accuracy as low as ~17.2% in a 2018 study (Salberg & Larsen) — illustrates how hard biogenic look-alikes are; this paper's own aggregate metric not extracted
C2	Dabboor et al. — emulsion vs. plant-oil discrimination	2018	Biogenic (plant oil) vs. emulsion	Compact-pol features	Kappa = 0.599
C3	Gao et al. — 4-class CP-SAR scenario	2022	Crude oil vs. emulsion (4-class)	CP-SAR features	Kappa = 0.6599
C4	Novel DL method for marine oil-spill detection	2022	Low wind, biogenic slicks, rain cells, grease/frazil ice, current shears, wind shadows, upwelling, internal waves	Reviews polarimetric (quad-pol) features: co-pol phase difference, degree of polarization, conformity coefficient, entropy	Notes quad-pol swath (25–50 km) is too narrow for operational monitoring vs. single/dual-pol (300–500 km) — an important trade-off for your design
C5	Brazil 2019 spill — RIOSS system evaluation vs. look-alikes	2022	Rain cells, biofilms, low-wind	Random Forest on shape/texture PDFs	RIOSS misclassified one biofilm slick as a low-wind region with oil spots; otherwise captured false positives correctly
C6	Oil Spill Detection by SAR Images: Dark Formation Detection review	—	All major look-alike types	Rule-based photo-interpretation criteria (shape, seasonality, morphology, sheltering)	Qualitative decision criteria used by human photo-interpreters — useful as an explainability rule-layer in your pipeline
C7	ViR-SC ensemble (see B6)	2026	Biogenic films, surfactant layers, internal waves, low-wind zones	Denoising autoencoder + hybrid ensemble, explicitly built to reduce look-alike confusion	See B6 metrics

Practical implication: classical polarimetric/texture features (contrast, shape elongation, PDFs of backscatter) still materially help distinguish look-alikes even in a deep-learning pipeline — the strongest recent systems (B6, B8) fuse DL segmentation with classical-feature classifiers as a look-alike rejection stage rather than relying on segmentation alone.

4. Category D — AIS-Based Vessel Correlation & Source Attribution

#	Study	Year	Region	Method	Key finding	Public?
D1	Tracing illegal oil discharges from vessels using SAR and AIS	2021	Bohai Sea, China	Correlates SAR-detected vessel position + slick geometry with AIS; notes discharged slicks appear as linear dark streaks with a bright tip (the vessel)	Discharged oil may persist 2–12 h on the surface, constraining the AIS look-back window	Not open
D2	A new ship-tracing technology from oil spills based on multi-source data	2024	Not specified	Converts oil-spill polygon into trajectory points; **bidirectional drift model** (forward + backward) matched against AIS trajectory points to rank suspect vessels	Explicitly designed for the case where the ship is no longer touching the slick in the image (common due to satellite revisit gaps) — a probability-based ranking model	Not open
	Chang et al. —	2024		Spatio-temporal correlation between slick	Reported to significantly improve source-attribution	

D3	joint SAR–AIS framework	(as cited)	Not specified	and AIS tracks	accuracy (exact metric not in excerpt)	Not reported
D4	Oil Spilling Detection using AIS + Satellite Datasets	2025	India (academic project)	Isolation Forest (AIS anomaly detection) + SVM (satellite image classification), web dashboard	Framed as a real-time-capable dual-layer system; no benchmark accuracy extracted	Student project; code availability not confirmed
D5	Oil-trajectory analysis for surveillance planning, Mediterranean MPAs	2023	Mediterranean Sea	Uses CMEMS currents + EMODnet AIS + MAPAMED protected-area boundaries + Sentinel-1 to identify where to concentrate surveillance	Not a per-vessel attribution model but a surveillance-prioritization layer — relevant for your alerting/patrol-routing module	Data sources are public (CMEMS, EMODnet)
D6	CLS — Kharg Island oil-spill & maritime-intelligence case study	2026	Persian Gulf (Kharg Island)	Combines Sentinel-1/2 detection, AIS data fusion, and drift-model comparison (tidal-current-only hypothesis tested against observed slick extent)	Demonstrates operational workflow: detect → hypothesize source (terminal) → simulate forward drift → compare to observation	Commercial service, methodology described publicly
D7	CleanSeaNet (EMSA) vessel correlation	ongoing	EU waters	Detected vessels in SAR images correlated with "vessel position reports from other EU sources" (i.e., AIS/VMS) to increase likelihood of correct polluter identification; alerts issued to coastal states within 20 minutes of image acquisition	Operational EU system since ~2007	Government system, not public code

Key structural insight across D1–D7: none of the papers retrieved report a hard, validated "distance-to-culprit-vessel" accuracy metric (e.g., in km or %) as a general benchmark — this is a genuine, reported gap (see Section 18), and most published methods are demonstrated on 1–4 case studies rather than large statistically validated test sets.

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5. Category E — Backward Trajectory / Hindcasting Models

#	System	Developer	Type	Data inputs	Status
E1	**GNOME** (General NOAA Operational Modeling Environment)	NOAA Office of Response & Restoration	Lagrangian particle trajectory model; supports forward *and* backward runs	Wind (ECMWF ~1/8°, or NOAA-GFS ~0.25°), currents, oil "weathering" parameters	Free, open, desktop + used operationally by NOAA; includes **TAP/WebTAP** for statistical (many-run) contingency planning
E2	**OpenDrift / OpenOil**	MET Norway (Dagestad et al.)	Open-source Python Lagrangian framework; OpenOil module does 3-D oil drift + weathering (evaporation, emulsification, entrainment via Tkalich & Chan 2002 or Li et al. 2017 parameterizations); weathering code adapted from NOAA PyGnome/OilLibrary	Any current/wind reader (CMEMS, ECMWF, etc.)	**Fully open source (GPLv2), GitHub-hosted, actively maintained** — strong MVP candidate for both forward drift and (by reversing time) approximate backward hindcasting
E3	MEDSLIK-II	Academic/operational (Mediterranean)	3-D oil-spill/weathering model	Wind (SKIRON ~5 km), currents	Widely used research/operational model in Mediterranean case studies
E4	BLOSOM	US agencies	4-D fate/transport model, supports seabed-origin spills	Currents, diffusion, wind advection	Compared against GNOME in Duran et al. 2018
E5	OSCAR, SIMAP, OILMAP, MOHID, EUROSPILL, OILTRANS	Various commercial/academic	Oil fate & transport models (forward, some support backward runs)	Currents, wind, waves	Mix of commercial and research licenses

E6	INCOIS Online Oil-Spill Advisory (India) — see Section 8	INCOIS	Trajectory prediction system (forward, contingency-style) using Indian Ocean circulation + atmospheric models + GIS	High-resolution ocean currents (9.25 km Indian Ocean-wide; 2.3 km on the west coast)	Operational for Indian Coast Guard since 2016 (v3.0)
E7	GNOME validated against Indian high-resolution currents	Prasad, Francis, Balakrishnan Nair, Shenoi, Vijayalakshmi (INCOIS)	Peer-reviewed	High-resolution ocean-current fields	Journal of Operational Oceanography, 2019 — directly relevant precedent for an India-focused MVP

On "backward hindcasting" specifically: none of the operational models (GNOME, OpenDrift, MEDSLIK-II) are architecturally different for backward vs. forward — they are the *same* Lagrangian particle-advection engine run with reversed time and (ideally) reversed/adjoint diffusion. This is standard practice in the oil-spill-response community (source-point back-tracking), but uncertainty grows the further back in time you go, and turbulent diffusion is not perfectly reversible — treat backward runs as a probability cloud of candidate origins, not a single point.

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6. Category F — Forward Oil-Spill Drift Prediction

Covered largely by the same engines as Section 5 (GNOME, OpenOil, MEDSLIK-II, OSCAR/SIMAP/OILMAP family). Additional ML-specific forward-prediction items:

#	Study	Method	Notes
F1	Predictive analytics for oil-spill trajectories: a machine-learning approach (Mao et al., 2023)	ML regression/hybrid approach	Cited secondhand in D4; underlying paper not independently verified in this pass — **treat as "reported by a secondary source," verify before citing further**
F2	Kharg Island case (CLS)	Physics-based drift model (tidal currents) validated against SAR-observed slick extent	Demonstrates a "simulate-then-validate-against-imagery" loop directly applicable to a forward-consistency score in your attribution formula
F3	Ocean-wave-trajectory forecasting system, eastern Baltic (validated with drifting buoys)	OpenDrift/OpenOil forced by NEMO + SWAN + ECMWF	Demonstrates a full met-ocean-forced operational pipeline suitable as an architecture template

State of the art: pure physics-based Lagrangian models (GNOME/OpenOil/MEDSLIK-II) remain the dominant, most validated approach for forward drift. Machine-learning "surrogate" or "correction" models for drift trajectories are an active but much less mature research area than SAR detection — this is a genuine novelty opportunity (see Section 18).

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7. Category G — Integrated Satellite–AIS–Oceanographic Systems

#	System	Operator	Integration level	Notes
G1	**CleanSeaNet**	EMSA (EU)	Satellite detection (SAR + optical) → vessel correlation (AIS/VMS) → GIS alert to coastal state within 20 min → potential ship-inspection follow-up	The most complete **operational government reference system**; free to EU/EEA authorities; not public API for outsiders
G2	KSAT Oil Spill Detection Service	Kongsberg Satellite Services (Norway)	Satellite detection + analyst review, "fully analysed results globally every day"	Commercial, >20 years operating experience
G3	CLS Maritime Surveillance / Oil Spill Monitoring	CLS (France)	Detection + drift + "potential polluter" identification	Commercial; Kharg Island case study shows the AIS+drift+SAR loop end-to-end
G4	ORBITAL (OrbitalEOS)	Orbital	AI + satellite, claims 95% detection accuracy (vendor-reported, not independently verified)	Commercial; treat vendor accuracy claims cautiously — "not independently reported"
	INCOIS Online			Government-operational; forward trajectory + GIS, **no AIS-**

G5	Oil-Spill Advisory (OOSA v3.0)	INCOIS (India)	Trajectory model + GCM + atmospheric model + GIS, eco-sensitive-zone and fishing-zone overlays	**based attribution component identified in available material** — a documented gap for an Indian-context product

Integration-completeness observation (used again in Section 23): of all systems reviewed, **CleanSeaNet is the only one that visibly closes the full loop** — detection → AIS correlation → alert → follow-up. Backward-hindcast-based *automated* vessel attribution scoring (rather than manual analyst correlation) does not appear as a public, named operational product in any source retrieved.

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8. Category H — Indian Research & Operational Systems

#	Item	Agency	Description	Access
H1	**Online Oil Spill Advisory (OOSA), v3.0**	INCOIS (Ministry of Earth Sciences), with Indian Coast Guard	Oil-spill trajectory prediction using oil-spill model + general circulation model + atmospheric model + GIS; identifies eco-sensitive zones, potential fishing zones, fishing-avoidance zones; incorporates 9.25 km Indian-Ocean-wide and 2.3 km west-coast current grids; inaugurated 2016 at the 21st NOSDCP meeting	Advisory service for Coast Guard / stakeholders, not a public API
H2	High-resolution ocean-current-based trajectory validation	Prasad, Francis, Balakrishnan Nair, Shenoi, Vijayalakshmi (INCOIS), published 2019	Peer-reviewed validation of trajectory prediction using high-resolution Indian Ocean currents	Journal of Operational Oceanography
H3	RS-GIS based Operational Monitoring of Indian Maritime and Environs	Academic (India)	Conceptual framework proposing RISAT/Oceansat SAR + optical + ISRO Bhuvan/NICES geoportals + INCOIS web services + IRNSS positioning for an integrated Indian oil-spill monitoring architecture	Proposal/review paper, not an operating system as far as retrieved sources show
H4	**MOSDAC** (Meteorological & Oceanographic Satellite Data Archival Centre)	ISRO / Space Applications Centre (SAC)	Data repository for INSAT, Megha-Tropiques, Oceansat, SCATSAT-1 winds, and other Indian meteorological/oceanographic products; also issues cyclone-track and weather-alert products	Free registration-based access
H5	AIS + Satellite dual-layer oil-spill detection (student project)	ISL Engineering College, Hyderabad	Isolation Forest (AIS anomaly) + SVM (satellite classification), web interface	Academic project (2025); illustrates a low-cost architecture pattern feasible for an India-based MVP

Gap identified for India: INCOIS's OOSA is a strong, validated **forward-drift/contingency** system, but the retrieved material shows **no public evidence of an automated AIS-based vessel-attribution module** integrated with it, nor of deep-learning SAR segmentation being operationally coupled to it. This is precisely the gap your application could fill for the Indian EEZ (Exclusive Economic Zone), including high-traffic corridors (Gulf of Kutch/Kambhat are explicitly mentioned as oil-spill-prone training scenarios in INCOIS material).

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9. Category I — Open-Source Datasets, APIs & GitHub Projects

#	Resource	Type	Content	Access	Link
I1	**Krestenitis oil-spill dataset (MKLab/CERTH)**	Dataset	1,112 Sentinel-1 VV images, 5-class segmentation masks (sea, oil, look-alike, ship, land); the de facto SAR oil-spill benchmark	On request via institutional email to MKLab/CERTH — **not a one-click download**	Paper, TorchGeo tracking issue
I2	Kaggle "oil-spill" dataset (nabilsherif)	Dataset	Sentinel-1 SAR imagery annotated for oil spill (used directly in an open GitHub project)	Free, Kaggle account required	referenced in Harsha0112/Oil-Spill-Detection

I3	Sentinel-1 SAR Oil Spill Image Dataset, Part I	Dataset	2048×2048×2 Sigma0 images (dB) + TIFF ground-truth masks	Listed in the community-maintained satellite-image-deep-learning datasets index	GitHub index
I4	OPD v1.0.1 — Offshore Platform Dataset	Dataset	Sentinel-1-derived geolocations, lifespans, spatial attributes of offshore oil/gas platforms — directly useful as a "candidate source location" prior layer	Free	GitHub index
I5	**OpenDrift / OpenOil**	Software (GitHub)	Full Lagrangian ocean-trajectory + oil-weathering engine, Python, GPLv2	Free, open source	github.com/OpenDrift/opendrift
I6	NOAA **GNOME**, **TAP/WebTAP**, **GOODS**, **ADIOS/OilLibrary**	Software suite	Trajectory modeling, contingency planning, current/wind data access helper, oil-weathering database	Free, downloadable	NOAA GNOME suite
I7	Harsha0112/Oil-Spill-Detection (GitHub)	Code	U-Net + DeepLabV3 training pipeline + AIS-based vessel anomaly ROI generation, uses MarineCadastre AIS data	Open source	GitHub
I8	oceanhackweek/ohw23_proj_oil ("Project SisMOM")	Code	Notebook to classify oil-slick vs. look-alike from cropped SAR chip statistics	Open source	GitHub
I9	juancarloscno/oil-spill-SAR	Code	Oil-spill detection pipeline for Ecuadorian EEZ, includes SNAP-based preprocessing scripts, Docker/Enroot containers	Open source	GitHub
I10	Random-Forest open-source oil-spill detector	Code + trained models	Two RF classifiers with interpretable feature set specifically tuned to reject look-alikes	Open source	MDPI paper
I11	**MarineCadastre.gov AccessAIS**	AIS dataset (US)	Historical U.S. coastal AIS data, free bulk download	Free	referenced in Harsha0112 repo
I12	**Copernicus Data Space Ecosystem**	Sentinel-1/2 data + API	Full free access to Sentinel-1 SAR (GRD/SLC) and Sentinel-2 optical; STAC/OData API	Free (registration)	dataspace.copernicus.eu
I13	**Copernicus Marine Service (CMEMS)**	Ocean current/SST/wave API	Physical ocean-model reanalysis and forecast currents used directly in several papers above (e.g., Mediterranean surveillance study)	Free (registration)	referenced in Ciappa 2023
I14	**EMODnet**	Human-activity / AIS-derived vessel-density data	Used as the AIS source in the Mediterranean surveillance-prioritization study	Free	referenced in Ciappa 2023

10. Category J — Commercial Oil-Spill Monitoring & Attribution Systems

#	Vendor	Offering	Notes
J1	**EMSA CleanSeaNet**	Government (not commercial) satellite oil-spill + vessel detection for EU states	Reference "gold standard" workflow — see G1

J2	**KSAT** (Kongsberg Satellite Services)	Global oil-spill detection-as-a-service, human-analyst-reviewed	20+ years operating, cited as the source of a large curated training dataset (713 products / 2,093 events) used in academic large-scale detection research
J3	**CLS** (Collecte Localisation Satellites)	Satellite detection + AIS fusion + drift modelling + "potential polluter" identification	Publicly documented case study (Kharg Island) shows their end-to-end workflow
J4	**Viridien** (formerly CGG)	Oil-spill detection & monitoring for offshore/coastal risk	Commercial geoscience/satellite-mapping provider
J5	**Orbital (OrbitalEOS)**	AI-based near-real-time detection, vendor claims ~95% accuracy	Vendor-reported metric — **not independently verified**, treat cautiously
J6	**OSRL** (Oil Spill Response Ltd.) satellite remote sensing (via MDA partnership)	SAR + optical for spill detection, vessel tracking, GIS-ready output	Industry-funded not-for-profit response cooperative
J7	**Spire Global** (acquired exactEarth) / **Orbcomm** / **Planet Labs**	Satellite-AIS (S-AIS) data feeds; Spire also offers "Enhanced Satellite AIS," "Port Events"	Commercial AIS data providers; relevant as your paid-AIS upgrade path if free AIS coverage is insufficient
J8	**MarineTraffic**, **Kpler**, **Starboard Maritime Intelligence**	Vessel-tracking / dark-vessel & spoofing-detection platforms	Kpler reported identifying 261 AIS-spoofing vessels between Jan 2024–Jul 2025; Starboard focuses on illegal-fishing/dark-vessel detection for governments

11. Machine-Learning Method Guide (what each model is good for)

Model family	Best suited for	Evidence from this review
U-Net & variants (Dual-Attention U-Net, U-Net++)	Pixel-level oil-slick segmentation, especially with limited labeled data; encoder-decoder captures multi-scale slick shapes	Standard baseline in nearly every SAR study (A1, B1, B4, B6)
DeepLabv3+	State-of-the-art boundary-quality segmentation when a moderate-to-large labeled dataset exists; atrous conv + ASPP handles multi-scale slicks well	Outperformed U-Net/LinkNet/PSPNet/DeepLabv2 in the original benchmark (A1); best-of-class result (IoU 91.6%) on a curated 35-scene set (B2)
YOLO / other object detectors	Fast slick or vessel *localization* (bounding boxes) rather than exact pixel boundary — useful as a first-pass triage stage before full segmentation, or for vessel detection in the same SAR scene	Referenced conceptually in compositional detector + SAM work (B7); not benchmarked with hard numbers in sources retrieved
Mask R-CNN	Instance segmentation — when you need to separate *multiple distinct slicks or vessels* in one scene, not just a binary oil/non-oil mask	Explicitly described as detection + segmentation combined (B7 context)
CNN + transfer learning	Classification stage (oil vs. non-oil chip) before segmentation; transfer learning from ImageNet/other SAR corpora reduces the labeled-data burden, which matters given how small most oil-spill training sets are (35–1,112 images)	A4 (23-layer CNN classifier), B1
Vision Transformers (ViT)	Best combined with CNNs in a hybrid ensemble for global-context modeling of long, thin slick shapes; on their own, ViTs need more data than most oil-spill datasets provide	B6 (ViT as one arm of the ViR-SC ensemble)
Random Forest, SVM, XGBoost	Classical, interpretable classifiers — strong choice specifically for **look-alike rejection** using hand-engineered polarimetric/texture/shape features, and for AIS-based anomaly scoring where interpretability matters for legal/evidentiary use	C1–C3, I10 (open-source RF detector), D4 (Isolation Forest for AIS anomalies), D5/C5 (RF on shape/texture PDFs)
LSTM, GRU, Temporal CNN, Transformer (for AIS trajectories)	Vessel trajectory prediction/forecasting, anomaly detection (course/speed deviation), and filling AIS gaps	Multiple studies D-adjacent: FRA-LSTM (bi-directional LSTM fusion), LSTM-DCross (handles GPS-failure gaps), CNN+DNN+LSTM+GRU South Korean study, AIS-LLM (multi-task transformer-style, reports ADE 0.43/FDE 0.91 on Piraeus AIS data)
Graph Neural		Vessel trajectory abnormal-behavior detection via graph-attention

Networks (GNN) for vessel tracks	Modeling *interactions between multiple vessels* (traffic-graph structure) for anomaly/collision-risk detection, not just single-vessel trajectory extrapolation	prediction/reconstruction network (Section on AIS trajectory search); GTRA (graph-based traffic representation + evidential DL)
Data fusion / multimodal models	Combining SAR imagery + AIS + oceanographic fields in one scoring or learning pipeline	D2 (bidirectional drift + AIS point matching), D6 (CLS operational fusion), B6 (DL + classical ensemble)
Physics-informed ML	Constraining an ML drift-correction model with known advection-diffusion physics so predictions stay physically plausible; the field currently leans on **pure physics models (GNOME/OpenOil/MEDSLIK-II)** rather than physics-informed neural nets **— explicit ML-physics hybrids for oil drift were not found as a mature, named system** in sources retrieved. This is a genuine research gap, not just an oversight of this review.	Absence itself is evidence — see Section 18
Ensemble & uncertainty-estimation methods	Combining independently-trained models (DL segmentation + classical classifier) to reduce false positives, and quantifying confidence for downstream attribution scoring	B6 (explicit multi-model ensemble); NOAA TAP explicitly runs "thousands of GNOME runs" to build a statistical spread — the closest analogue to formal uncertainty quantification found in this review

12. Data Feasibility Assessment

Data need	Source	Cost / access tier	Key constraint
SAR imagery	**Sentinel-1** (ESA/Copernicus)	**Free**, registration via Copernicus Data Space Ecosystem	As of 2026 the constellation is mid-transition: Sentinel-1A is being phased out (terminated ~29 June 2026) in favor of Sentinel-1C+1D, restoring the nominal **6-day revisit** cycle (temporarily as short as 1 day during the overlap window in April–June 2026). Build your ingestion layer against the STAC/OData API, not against a specific satellite ID, so it survives future constellation changes.
Optical confirmation	**Sentinel-2** MSI	Free	Cloud cover and daylight-only acquisition sharply limit usability for near-real-time detection — best used as a *secondary confirmation* layer, not the primary detector, consistent with how every reviewed operational system (CleanSeaNet, KSAT, CLS) treats it.
Free/low-cost AIS	**MarineCadastre.gov (US)**, **EMODnet (EU)**, national VTS feeds	Free (bulk historical, some near-real-time)	Coverage limited to areas with terrestrial receivers unless satellite-AIS is added; message gaps, spoofing, and deliberately disabled transponders ("dark vessels") are a documented, persistent limitation across every AIS source reviewed.
Commercial/satellite AIS	**Spire Maritime (absorbed exactEarth), Orbcomm, Planet Labs, MarineTraffic, Kpler**	Paid subscription/API, tiered by coverage and refresh rate	Extends coverage to open ocean and improves refresh rate to minutes–hours, but does **not** solve spoofing (Kpler reported 261 spoofed-position vessels identified Jan 2024–Jul 2025) or deliberate AIS shutdown; some providers now offer forensic/chain-of-custody data provenance specifically to counter this.
Ocean currents	**Copernicus Marine Service (CMEMS)**	Free (registration)	Global + regional reanalysis and forecast products; used directly in the Mediterranean surveillance study and is the natural default for a India-adjacent or global MVP.
Wind / met data	**ECMWF (IFS, free tier via Copernicus/CDS), NOAA GFS**	Free (registration); higher-resolution/faster-refresh commercial tiers exist	ECMWF ~1/8° and NOAA-GFS ~0.25° resolutions were both reported as adequate GNOME-forcing inputs in the literature (Section 5).
Bathymetry	**GEBCO**	Free	Standard global bathymetric grid; used for coastal/seabed-spill scenarios and stranding models.
Indian ocean/weather data	**MOSDAC** (ISRO/SAC) — INSAT, Oceansat, SCATSAT-1 winds	Free registration	Freeware license per MOSDAC's own listing; good local complement to CMEMS/ECMWF for an India-focused deployment.
		Advisory service to authorities,	

Indian oil-spill trajectory reference	**INCOIS OOSA**	not an open API as far as retrieved sources show	Consider this a validation reference / potential government partnership target rather than a data source you can freely ingest.

Licensing/attribution notes: Copernicus data (Sentinel-1/2) and Copernicus Marine Service data are released under free, open EU licensing that requires attribution but permits commercial use — verify current terms on the Copernicus Data Space Ecosystem site before shipping, since these terms can be updated. AIS data resold by commercial vendors typically carries redistribution and rate-limit restrictions in their commercial contracts — read each vendor's terms of service directly; this review does not have visibility into current contract specifics for Spire/Orbcomm/Kpler and you should not assume free-tier terms extend to production use.

Bottom line on feasibility: a fully free-data MVP is realistic for **detection + forward drift + basic attribution scoring** (Sentinel-1 + CMEMS + ECMWF/GFS + MarineCadastre/EMODnet AIS + OpenDrift/GNOME), but **attribution confidence in open ocean or near sanctioned/dark-fleet activity will be structurally limited** by free-AIS coverage gaps — this is a data-availability ceiling, not a modeling problem, and no amount of ML sophistication removes it.

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13. End-to-End Architecture

1. **Data ingestion layer** — scheduled pullers for Sentinel-1 (STAC API), AIS (free + optional commercial), CMEMS currents, ECMWF/GFS wind/wave, GEBCO bathymetry; message-queue (e.g., Kafka) to decouple ingestion from processing.
2. **Satellite-image preprocessing** — radiometric calibration, speckle filtering (Lee/Refined Lee or the denoising-autoencoder approach from B6), land/ship masking, geocoding to a common grid.
3. **Oil-spill detection & segmentation module** — DeepLabv3+ or U-Net (start with a transfer-learned backbone given small training sets), output per-pixel class + confidence map.
4. **Look-alike rejection module** — secondary classical-feature classifier (RF/SVM/XGBoost on polarimetric, shape, and contextual features) fed with wind-speed context (spills are unlikely at <2–3 m/s or >10 m/s per the Suez Canal study); flags biogenic/rain/low-wind candidates for human review rather than silent auto-reject.
5. **AIS data cleaning & map matching** — deduplicate, interpolate short gaps, flag implausible speed/position jumps (a proxy for spoofing), snap to shipping-lane/port/terminal reference layers.
6. **Candidate-vessel search algorithm** — spatio-temporal window query (e.g., ±160 km per the shoreline-proximity prior from B7, ± several hours per the 2–12 h surface-persistence window from D1) around each detected slick.
7. **Forward drift simulation** — OpenDrift/OpenOil (or GNOME) seeded at each candidate origin (vessel position at time T), forced by CMEMS + ECMWF/GFS.
8. **Backward trajectory / source-location estimation** — same engine run in reverse from the observed slick polygon, producing a probability cloud of origin points/times rather than a single answer.
9. **Spatio-temporal vessel-to-spill scoring model** — combine forward-consistency, backward-consistency, and AIS-track proximity into the attribution formula (Section 14).
10. **Vessel attribution & confidence-ranking module** — rank candidate vessels, propagate uncertainty from both the segmentation confidence and the drift-model ensemble spread (TAP-style multi-run approach).
11. **GIS map & visualization layer** — web map (e.g., Leaflet/Mapbox/OpenLayers) showing slick polygon, drift cloud, candidate vessel tracks, eco-sensitive zones (INCOIS-style overlays).
12. **Alert & notification system** — threshold-based push (email/SMS/webhook) to responders, mirroring CleanSeaNet's ~20-minute alert SLA as a design target.
13. **Database & cloud architecture** — object storage for raw imagery, a spatio-temporal database (PostGIS/TimescaleDB) for AIS + detections + drift outputs, containerized inference services (Docker/Kubernetes) for horizontal scaling during high-detection periods.
14. **Model training, validation & monitoring pipeline** — versioned datasets, held-out regional test sets (don't just reuse Krestenitis splits — build a local/regional validation set given how small and geographically narrow most public benchmarks are), drift/data-quality monitoring in production.
15. **Human-in-the-loop review workflow** — every reviewed operational system (CleanSeaNet, KSAT, CLS) keeps a human analyst in the loop for confirmation before action is taken; design your UI around fast analyst triage (auto-ranked candidate list, one-click accept/reject) rather than full automation.
16. **Explainability & evidence-report generation** — auto-generate a PDF/GIS package per incident: detection confidence map, look-alike-rejection reasoning, drift-model run parameters and spread, AIS track overlay, and the attribution score breakdown — this is what would actually be submitted as evidence, so make every score component traceable to its inputs.

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14. Proposed Source-Attribution Scoring Formula

No single, standardized attribution formula exists in the literature reviewed (this is a genuine gap — see Section 18). The formula below synthesizes the scoring factors your prompt specified, weighted according to the qualitative importance each factor was given across the sources above (e.g., the 2–12 h persistence window from D1, the 160 km shoreline-proximity prior from B7, the bidirectional-drift approach from D2).

```
AttributionScore(v, s) =
    w1 · f_distance(d_v,s)              # decays with vessel-to-slick distance at detection time
  + w2 · f_time(Δt_v,s)                 # decays with time gap vs. the 2–12h oil-persistence window
  + w3 · f_kinematics(speed_v, heading_v, wind, current)   # consistency of vessel motion vs. observed slick elongation/orientation
  + w4 · f_backward(P_backward(v))      # likelihood vessel position falls inside the backward-hindcast probability cloud
  + w5 · f_forward(P_forward(v))        # likelihood forward-simulated drift from vessel's track matches observed slick polygon
  + w6 · f_vesseltype(type_v, cargo_v)  # tankers/cargo ships up-weighted over passenger/fishing vessels for oil source plausibility
  + w7 · f_proximity_infra(d_to_lane, d_to_port, d_to_platform, d_to_pipeline)  # proximity to known oil-handling infrastructure
  + w8 · f_spillprofile(size, shape, estimated_age)   # e.g., long linear streak = discharge-in-transit; static blob = leak/grounding
  + w9 · f_AIS_quality(completeness, gap_ratio, spoof_flag)  # down-weights vessels with poor/suspicious AIS records (which paradoxicall
  - u1 · Uncertainty(detection_confidence, drift_model_spread)   # subtract, don't hide, uncertainty from the final score
```

Design notes:

- Weights (w1…w9) should be **learned or calibrated per region** (traffic density, typical AIS quality, and current/wind regime all differ, e.g. Suez Canal vs. open Indian Ocean) rather than fixed globally.
- Keep the **AIS-quality term separate from a "suspicion" term**: a vessel with poor/absent AIS is not automatically more likely to be the polluter, but it is a case the system should escalate for manual review rather than silently score low or high.
- Report **uncertainty as a first-class output alongside the score**, not just a single number — this mirrors NOAA's TAP approach of running many drift simulations to get a spread rather than one deterministic path.
- This formula is a design proposal from first principles combining your requirements with patterns found in the literature — it has **not** been benchmarked in this review, and should be validated against known historical spill/vessel-attribution cases (e.g., known enforcement outcomes) before being trusted operationally.

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15. Distinguishing Spill Causes

Cause	Distinguishing signals from this review
Intentional operational discharge	Long, thin, linear dark streak with a bright tip (the vessel) in the SAR image (D1); vessel typically in transit (steady speed/heading) rather than stopped; often near but not exactly on shipping lanes to avoid easy detection
Accidental collision or grounding	Spill origin coincides with a stationary or erratically-moving AIS track near shoals/reefs/other vessels; often followed by a distress signal or an abrupt AIS gap
Tanker or cargo-vessel leakage	Spill tracks a vessel's route over time (multiple detections aligned with one AIS track) rather than a single discharge event; may show gradual size growth
Offshore-platform or pipeline leakage	Spill origin is static and repeats near the same coordinates over multiple satellite passes; matches the fixed locations in an offshore-platform dataset such as OPD (I4)
Natural seepage	Recurring slicks at the same coordinates over years, often in known seep provinces, no correlated AIS track nearby
Biogenic slicks	Backscatter/texture signatures statistically distinct from mineral oil in polarimetric feature space, though with substantial overlap (Kappa ≈0.6 in cited classifiers — Section 3); often seasonal (e.g., algal bloom periods)
Low-wind / rain-related SAR artefacts	Co-located with independently-sourced low-wind-speed fields (<2–3 m/s) or rain-cell radar/weather data; lack the sharp, persistent-shape boundary of true oil
Vessels with missing/manipulated AIS	Detected vessel in SAR imagery has no corresponding AIS track, or the AIS track shows implausible jumps/duplicated MMSI — treat as a distinct "dark/spoofed vessel" investigative branch, cross-referenced against known spoofing/dark-fleet databases from commercial AIS vendors (Section 9, J8)

None of these distinctions are reported in the literature as solved by a single automated classifier — in every operational system reviewed (CleanSeaNet, KSAT, CLS), a **human analyst makes the final determination** using exactly this kind of contextual reasoning; treat automated cause-classification as a decision-support ranking, not a final verdict.

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16. Recommended MVP

MVP features (build first):

- Sentinel-1 ingestion + DeepLabv3+ or U-Net segmentation (transfer-learned on Krestenitis + fine-tuned on your target region)
- Classical-feature look-alike rejection stage (RF/SVM), flagged for human review rather than silently dropped
- Free/registered AIS ingestion (MarineCadastre/EMODnet, or a national feed if available) + basic cleaning/gap interpolation
- OpenDrift/OpenOil forward-drift simulation from candidate vessel positions
- Simple backward hindcast (reverse-time OpenDrift run) producing a probability cloud
- Basic attribution scoring (a reduced version of Section 14's formula, without the infrastructure-proximity/vessel-type layers initially)
- Web GIS map (slick + drift cloud + candidate vessels)
- Manual-review dashboard (human-in-the-loop, no auto-alerting yet)

Postpone to later phases:

- Commercial/satellite AIS integration (only add once free-AIS gaps are shown to materially hurt attribution accuracy in your pilot region)
- Full explainability/evidence-report PDF generation
- Real-time alerting/notification system
- Vessel-type/cargo enrichment and pipeline/platform-proximity layers
- Physics-informed or learned drift-correction models
- GNN-based multi-vessel interaction modeling

Datasets required for MVP: Krestenitis benchmark (request access early — it is not instant-download), Sentinel-1 GRD via Copernicus Data Space, MarineCadastre or EMODnet AIS, CMEMS currents, ECMWF/GFS wind.

Suggested stack: Python (PyTorch/TensorFlow for segmentation, scikit-learn for classical classifiers), OpenDrift (Python, drop-in), PostGIS/TimescaleDB, FastAPI backend, a Leaflet/Mapbox front end, containerized on a Kubernetes or single-VM Docker Compose setup for the MVP scale.

Suggested model choices: DeepLabv3+ (EfficientNet or ResNet backbone) as primary segmentation; Random Forest as the look-alike-rejection classifier for interpretability; OpenDrift/OpenOil as the physics engine — do **not** start with a from-scratch drift model; that is reinventing a decades-mature capability that NOAA/MET Norway already maintain and validate.

Minimum viable geographic area: one well-instrumented, moderate-traffic coastal zone with reasonable AIS terrestrial coverage — e.g., a stretch of the Indian west coast (Gulf of Kutch/Kambhat area is explicitly used in INCOIS training scenarios) or a similarly AIS-dense strait, rather than open ocean, to keep both Sentinel-1 revisit and AIS coverage adequate during the pilot.

Expected processing time: not reported in any source as a hard number for an integrated pipeline like yours; individual components suggest realistic near-real-time targets — CleanSeaNet issues alerts within ~20 minutes of image acquisition (a useful SLA target), while Sentinel-1 revisit itself is currently ~6 days per scene over a given point (shorter locally due to swath overlap) — your system's overall latency will be dominated by satellite revisit, not by your own compute, for detection, while AIS-based attribution can run near-real-time once a detection exists.

Expected accuracy and uncertainty: segmentation IoU in the reviewed literature spans roughly 53–96% depending on dataset size/curation and class definition — treat any accuracy number you build toward as sensitive to how you define/curate your local dataset. No source in this review reports a validated numeric accuracy figure for end-to-end vessel attribution — build your own hold-out evaluation using historical spill/vessel-identification cases with known outcomes, since the field itself does not yet provide a public benchmark for this specific task.

Estimated technical risks: (1) obtaining the Krestenitis dataset or an equivalent labeled corpus (access-gated, and even then narrow in scene diversity); (2) free-AIS coverage gaps in your pilot area undermining attribution confidence; (3) look-alike false positives/negatives, especially biogenic slicks (documented Kappa as low as ~0.6 in the literature); (4) backward-hindcast uncertainty compounding with time since spill; (5) legal/evidentiary standards for "attribution" that this review found no public technical benchmark for — you may need domain/legal-expert input to define what confidence threshold is actionable.

Evaluation metrics: IoU/mIoU/Dice/precision/recall for segmentation; precision/recall/F1 for look-alike classification; distance error (km) and time-to-detection for drift-model validation against known historical spill cases; top-k candidate-vessel accuracy (was the true polluter in your top-k ranked list?) for the attribution module, benchmarked against any cases with confirmed outcomes you can obtain (e.g., enforcement records).

Six-month roadmap (indicative)

Month	Milestone
1	Data access secured (Copernicus registration, Krestenitis dataset request submitted, AIS source selected); architecture skeleton (ingestion + storage) stood up

2	Segmentation model trained/fine-tuned on Krestenitis + regional Sentinel-1 scenes; baseline IoU established
3	Look-alike rejection classifier built and integrated; AIS cleaning/map-matching pipeline built
4	OpenDrift/OpenOil forward + backward runs integrated; candidate-vessel search implemented
5	Attribution scoring v1 implemented; GIS map + review dashboard built; internal testing on historical spill cases
6	Pilot-region field validation, metrics collected against Section 16 evaluation criteria, roadmap for Phase 2 (postponed features) drafted

17. Comparison Table (≥15 entries, cross-category)

#	Name	Year	Type	Region	Key tech	Metric (verbatim from source or "not reported")	Public
1	Krestenitis et al. benchmark	2019	Peer-reviewed	EU (EMSA-sourced)	U-Net/DeepLabv3+/etc.	DeepLabv3+ best; oil-class IoU ≈53–54% (per later comparative citation)	Dataset on-request
2	Ma et al. improved DeepLabv3+	2022	Peer-reviewed	Not specified	DeepLabv3+ variant	IoU 91.58% (35-scene curated set)	Not reported
3	Zhu et al. CBD-Net	2022	Peer-reviewed	Not specified	Custom CNN	Precision 84.2%/91.2% (PALSAR/S1)	Not reported
4	Suez Canal DeepLabv3+ study	2025	Peer-reviewed	Egypt	DeepLabv3+ (EfficientNet)	Accuracy 98.14%, MIoU 0.7872	Not reported
5	Trujillo-Acatitla et al. two-step DL	2024	Peer-reviewed	Mexico	CNN + U-Net	99% classification acc.; 99% seg. acc., 96% IoU	Not reported
6	ViR-SC hybrid ensemble	2026	Peer-reviewed	Not specified	CNN+ResNet18+ViT+RF+SVM+U-Net++	IoU 0.78, Dice 0.86, pixel acc. 93.5%	Not reported
7	KSAT large-scale detection study	2020	Peer-reviewed (arXiv)	Global (North Sea/Barents)	Segmentation CNN	713 products / 2,093 events, 2014–2018	Paper open; data not public
8	Gulf of Mexico operational monitoring	2024	Peer-reviewed	Mexico (Campeche Bank)	Multi-satellite + drifters/HF-radar	>1,500 images, 70+ reports, up to 669 km² spills	Not open-source
9	Open-source Random Forest detector	2021	Peer-reviewed + code	Not specified	RF classifiers	Not reported (numeric)	**Open source + trained models**
10	Compact-pol RF look-alike study	2025	Peer-reviewed	Not specified	RF + CP-SAR features	Cites prior Kappa 0.599–0.6599 for related work	Not reported
11	Bohai Sea SAR+AIS tracing	2021	Peer-reviewed	China	AIS correlation + SAR	Qualitative case study; 2–12h persistence window	Not open
12	Bidirectional drift + AIS ship-tracing	2024	Peer-reviewed	Not specified	Bidirectional drift model	Not reported (numeric)	Not open
13	Mediterranean MPA surveillance study	2023	Peer-reviewed	Mediterranean	CMEMS + EMODnet + Sentinel-1	Surveillance-prioritization output, not accuracy metric	Data sources public
14	CleanSeaNet	Ongoing (est. ~2007)	Government operational	EU	SAR/optical + AIS/VMS correlation	20-min alert SLA	Government system

15	KSAT Oil Spill Detection Service	Ongoing	Commercial	Global	SAR + analyst review	"20+ years" experience (vendor)	Commercial
16	CLS Maritime Surveillance	Ongoing	Commercial	Global (Kharg Island case shown)	SAR/optical + AIS + drift model	Case-study validated against tidal-current-only hypothesis	Commercial
17	GNOME / TAP	Ongoing (NOAA)	Government, open tool	US/global	Lagrangian trajectory model	Operational since 1990s; TAP runs 1000s of simulations for probability spread	**Free/open**
18	OpenDrift/OpenOil	Ongoing (MET Norway)	Open-source software	Global	Lagrangian + oil-weathering	Validated against drifting buoys (Baltic Sea study)	**Free/open (GitHub)**
19	INCOIS Online Oil Spill Advisory v3.0	2016–	Government operational	India / Indian Ocean	Trajectory + GCM + atmospheric model + GIS	9.25 km / 2.3 km current-grid resolution	Advisory service, not open API
20	AIS-LLM	2025	Peer-reviewed (arXiv)	Piraeus (test region)	Transformer/LLM-based multi-task AIS model	ADE 0.43 / FDE 0.91 (Piraeus), best-in-comparison	Not reported

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18. Research Gap Analysis

1. **No public, benchmarked end-to-end attribution metric.** Every source reviewed that touches AIS-based source attribution (D1–D7) demonstrates the concept on a handful of case studies; none reports a standardized precision/recall or "top-k candidate accuracy" the way segmentation research reports IoU. Building and publishing such a benchmark would itself be a contribution.
2. **Look-alike discrimination for biogenic slicks remains weak.** Reported Kappa scores of 0.6–0.66 for biogenic-vs-oil discrimination (Section 3) show this is nowhere near solved, despite being one of the oldest sub-problems in the field.
3. **No mature physics-informed ML for drift prediction.** The field still relies on pure physics engines (GNOME/OpenOil/MEDSLIK-II); ML "surrogate" drift models were not found as named, validated systems in this review.
4. **Public SAR oil-spill datasets are small and geographically narrow** (35–1,112 scenes typically, often concentrated in EU/Mediterranean/China coastal waters), which limits how well any pretrained model will generalize to, e.g., the Indian Ocean or tropical cyclone-affected regions without local fine-tuning.
5. **India-specific gap:** INCOIS's OOSA is a validated forward-drift/contingency system but shows no publicly documented AIS-based automated attribution layer or deep-learning SAR segmentation integration in the material retrieved.
6. **AIS spoofing/dark-vessel detection and oil-spill attribution are researched as separate literatures** (maritime-security AIS-anomaly research vs. oil-spill-response research) with limited visible cross-pollination in the sources retrieved — combining them is not yet standard practice.
7. **Uncertainty quantification is inconsistent.** Only NOAA's TAP (ensemble of many GNOME runs) offers a systematic multi-run uncertainty approach; most ML segmentation papers report a single point-accuracy number without confidence intervals.

19. Novelty Opportunities for Your Application

- A **formal, published attribution-scoring benchmark** (Section 14's formula, validated against historical confirmed-polluter cases) would fill Gap 1 directly and could become a citable contribution in its own right.
- **India-region fine-tuning + INCOIS-style integration**: coupling a DL segmentation + AIS-attribution pipeline to INCOIS's validated Indian Ocean current fields would address Gap 5, and the Gulf of Kutch/Kambhat corridor is already an INCOIS training scenario, making it a natural pilot site.
- **Explicit human-in-the-loop evidentiary reporting** — none of the reviewed academic systems produce a structured "evidence package" output; this is closer to an operational/legal-tech gap than a modeling gap, and is exactly what Section 13's module 16 targets.
- **Ensemble look-alike rejection with escalation-to-human routing** rather than binary reject — directly addresses the persistently weak Kappa scores in biogenic-slick discrimination (Gap 2) by making the system honest about where it is unreliable instead of forcing a hard classification.
- **Dark-vessel cross-referencing** — pairing your attribution pipeline with commercial dark-fleet/spoofing-detection feeds (Kpler, Starboard) to explicitly flag "no AIS match found — possible dark vessel" as its own actionable output category, addressing Gap 6.

20. Recommended System Architecture (summary)

See Section 13 for the full 16-module design. In one sentence: **Sentinel-1 → DeepLabv3+/U-Net segmentation → classical-feature look-alike rejection (human-reviewed) → AIS cleaning/map-matching → OpenDrift/OpenOil forward+backward runs → multi-factor attribution scoring with explicit uncertainty → GIS dashboard with human-in-the-loop confirmation → evidence-report export.**

21. Prioritized Datasets & APIs to Use

1. Copernicus Data Space Ecosystem (Sentinel-1/2) — free, foundational
2. Krestenitis MKLab/CERTH dataset — request access immediately, it is the standard benchmark
3. Copernicus Marine Service (CMEMS) — free ocean currents
4. ECMWF/CDS or NOAA GFS — free wind/wave forcing
5. MarineCadastre.gov / EMODnet AIS — free AIS to start
6. OpenDrift/OpenOil (GitHub) — free physics engine, do not rebuild
7. NOAA GNOME/TAP — free alternative/complementary physics engine and uncertainty methodology
8. GEBCO bathymetry — free
9. MOSDAC (ISRO) — free, for an India-specific deployment
10. Commercial satellite AIS (Spire/Orbcomm) — evaluate only after free-AIS gap analysis in your pilot region

22. Risk & Limitation Register

Risk	Category	Mitigation
Krestenitis dataset access delay (institutional email approval process)	Data access	Start the request in Month 1; build initial pipeline on the open Kaggle dataset (I2) in parallel
Free-AIS coverage gaps in open ocean	Data availability	Scope MVP to a coastal corridor; document the gap rather than overstating attribution confidence in open-ocean cases
AIS spoofing/dark vessels	Data integrity	Treat "no AIS match" as its own flagged category, not a silent failure
Biogenic/low-wind look-alike misclassification	Model accuracy	Route ambiguous cases to human review rather than auto-accept/reject
Sentinel-1 constellation transition (2026) changing revisit cadence	Operational	Build against the STAC/OData API abstractly, monitor ESA mission announcements
No standard attribution benchmark to validate against	Evaluation	Build your own hold-out set from historical confirmed cases; be explicit in any report that this is a novel, unvalidated-at-scale benchmark
Legal/evidentiary standard for "attribution" undefined	Deployment/adoption	Engage maritime-law or Coast Guard domain experts early to define what confidence threshold is operationally actionable
Vendor-reported accuracy claims (e.g., commercial "95% accuracy") unverifiable	Procurement	Do not rely on vendor marketing figures for your own evaluation baselines

23. Final Integration Table — Existing Systems vs. Components

System	Satellite detection	AIS tracking	Forward drift	Backward hindcast	Vessel attribution	GIS viz	Real-time alert
CleanSeaNet (EMSA)	✅	✅ (correlation)	—	—	Partial (manual/analyst)	✅	✅ (~20 min)
KSAT	✅	Not reported	—	—	—	Not reported	✅ (daily service)
CLS	✅	✅	✅ (case study)	—	Partial	Not reported	Not reported
INCOIS OOSA (India)	— (uses external SAR sources)	—	✅	—	—	✅	Advisory, not automated real-time
				✅ (reverse-		✅ (via	

NOAA GNOME/TAP	—	—	✅	time capable)	—	GOODS)	—
OpenDrift/OpenOil	—	—	✅	✅ (reverse-time capable)	—	— (bring your own viz)	—
Academic AIS+SAR studies (D1–D5)	✅ (varies)	✅	Partial	Partial (D2 bidirectional)	✅ (case-study level)	Varies	—
Your proposed application	✅	✅	✅	✅	✅ (novel, to be built)	✅	✅ (MVP-later)

Reading this table: the components you need already exist individually and are mostly free/open (Sentinel-1, OpenDrift, CMEMS). What does **not** exist as an integrated, publicly available system is the **combination** of automated DL-based detection + look-alike rejection + AIS attribution scoring + bidirectional drift + evidentiary GIS reporting in one pipeline — that combination is your product's real opportunity, not any single component in isolation.

Compiled from primary sources retrieved and read for this report. Where a specific number, dataset detail, or licensing term was not found in the sources retrieved, it is marked "not reported" rather than estimated — verify any load-bearing figure against its linked source before using it in a design decision or funding proposal.